

Digital Memory Assistant - Complete Project Vision & Master Blueprint

Executive Summary

Digital Memory Assistant is a desktop-first AI application that understands a user's digital life. It indexes local files, extracts meaning with AI, stores semantic embeddings and metadata, and lets users search using natural language instead of filenames.

Why This Project

Humans remember ideas and context, not filenames. Existing tools search locations; this product understands content and becomes a personal memory layer.

Project Type

AI-powered desktop application, semantic search engine, personal knowledge management platform, full-stack software product, NLP and information retrieval system.

Vision

Help people instantly find, understand and reconnect with everything they have ever learned, created or saved.

Goals

Offline-first, privacy-first, automatic organization, semantic search, AI chat, lifelong searchable knowledge base.

Target Users

Students, software engineers, researchers, professionals, creators and lifelong learners.

Why Desktop First

Local files remain private, indexing is faster, works offline and integrates directly with the user's computer.

User Journey

Install → Choose folders → Index files → AI processes content → Dashboard → Natural-language search → AI chat → Timeline → Background updates.

Workflow

File watcher → Indexer → Extractor → OCR/Transcription → Metadata → Summary → Tags → Embeddings → PostgreSQL → Qdrant → Hybrid search → AI answers.

Supported Data

PDF, DOCX, PPTX, TXT, Markdown, images, screenshots, audio, video, source code. Future: Gmail, Drive, GitHub, Calendar.

Architecture

React + Electron desktop UI, FastAPI backend, Indexing Engine, AI Engine, Search Engine, Analytics Service, Recommendation Service, PostgreSQL and Qdrant.

Algorithms

Trie, Inverted Index, ANN Search, Cosine Similarity, HashMap, LRU Cache, Priority Queue, Bloom Filter, Graph structures, Hybrid ranking.

Roadmap

Phase 1 Requirements, Phase 2 UI, Phase 3 Backend, Phase 4 Indexing, Phase 5 AI, Phase 6 Search, Phase 7 Desktop App, Phase 8 Testing, Phase 9 Deployment, Phase 10 Integrations.

Future Vision

Memory Health Dashboard, Knowledge Graph, Memory Map, Cloud Sync, Mobile Companion, Collaboration.

Guiding Principles

Privacy-first, local-first, modular architecture, scalable engineering, production-quality code, design before implementation.

Final Vision

Build a real product that acts as the user's second brain by understanding, organizing and retrieving knowledge across their digital workspace.

Major Features

- Universal Search
- Hybrid Search
- Semantic Search
- AI Chat
- OCR
- Speech-to-Text
- Automatic Summaries
- Automatic Tags
- Collections
- Timeline
- Dashboard
- Duplicate Detection
- Similar File Suggestions
- Voice Search
- Background Indexing
- Offline-first
- Learning Insights
- Memory Health Dashboard (planned)

Technology Stack

Component	Technology
Frontend	React + TypeScript + Tailwind CSS
Desktop	Electron
Backend	FastAPI
Database	PostgreSQL
Vector DB	Qdrant
AI	Sentence Transformers, Whisper, OCR, LLM
Workers	Celery + Redis
Deployment	Docker + GitHub Actions